AIRSTREET · USER REQUIREMENTS

AirStreet user requirements

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First user requirements, drawn from AST-PRB-001 Tables 1 and 2 (kit 1.9.1 upgrade)

AirStreet user requirements

What residents, city teams, schools, pole owners, researchers and installers need from a street air quality node, written from their side. They are drawn from the users and stakeholders in AST-PRB-001 (Tables 1 and 2) and are not yet validated with a partner; co-design sessions with the first partner (AST-DEC-001: a city or air agency with a street-level NO2 reference station) will revise them. Product requirements that meet these needs are in AST-PRS-001 (`03-requirements.md`) and the links are in the traceability matrix AST-RTM-001 (`03b-traceability.md`). Method: STANDARDS section 22.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Performance	Fine particles on my street	See PM2.5 on the street where I live, walk or send my children to school, close enough to a reference to trust	High	Residents and community groups (PRB Table 1)
UR-002	Performance	Traffic NO2 on my street	See NO2, the traffic gas, on the same street, close enough to a reference to trust	High	Residents and community groups; city teams (PRB Table 1)
UR-003	Performance	Rank streets with a stated uncertainty	Tell which streets are worse, and whether a street is above or below the legal limit, and know how sure that is	High	Municipal environment and transport teams (PRB Table 1)
UR-004	Performance	Data often enough for street maps	Readings often enough to see rush hours and daily patterns, and hourly values soon after the hour	High	City teams; researchers (PRB Table 1)
UR-005	Regulatory	A calibration others can check	A stated, repeatable calibration route, linked to each reading, that another group can audit	High	City teams; researchers; reference station operator (PRB Tables 1 and 2)
UR-006	Usability	Raw data to reprocess	Raw sensor signals and the calibration version, so data can be recalculated when a better model or a new collocation exists	Medium	Researchers and data platforms (PRB Table 1)
UR-007	Deployment and assembly	Fits existing poles without harm	Mount on existing street light and signal poles without drilling, without opening the pole and without its power	High	Street furniture owners (PRB Tables 1 and 2)
UR-008	Logistics	Small and light on the pole	A known, small load and wind area that the pole owner can approve	High	Street furniture owners (PRB Tables 1 and 2)
UR-009	Environment and durability	Works outdoors all year	Keep working in sun, rain, heat, cold, humidity, dust and traffic spray at street level	High	All users (PRB Table 1)
UR-010	Maintenance	Quick, simple service at height	Service at the pole in minutes with hand tools, with fiddly work done on the ground	Medium	Installers and service crews; city teams (PRB Table 1)
UR-011	Safety	Nothing personal collected	No images, audio or anything that identifies a person	High	Residents (PRB Table 1)
UR-012	Cost	Affordable for a district network	Low enough in parts and running cost for a community group or city to deploy tens of nodes	High	Community groups; city teams (PRB Table 1)
UR-013	Deployment and assembly	Own power supply	Run without mains, on its own solar and battery supply, through cloudy spells	High	Street furniture owners (PRB Table 2)
UR-014	Usability	Open and buildable		Medium	Schools and universities (PRB Table 1)

ID	Category	Title	Need	Priority	Source
			An open, documented design that a school or makerspace can build and repair		
UR-015	Safety	Safe to install and service	Installable and serviceable without danger from traffic, height, the pole's mains or the battery	High	Installers; pole owner (PRB Tables 1 and 2)
UR-016	Regulatory	Radio within the rules	Radio use within the band rules and the network's fair use where the node is deployed	Medium	Radio regulator and network operator (PRB Table 2)

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.